

Control and Coordination

The nervous system and neuron, reflex arc and human brain, movements in plants, plant hormones, and the endocrine system with feedback mechanism, with diagrams and practice questions.

Introduction

Our body responds to changes around us. When we touch a hot object we pull our hand back. Plants bend towards light. These responses need control and coordination. In animals this is done by the nervous system and the endocrine system. In plants it is done by hormones. This chapter explains how living things control their actions and coordinate with the environment.

Need for Control and Coordination

Living things respond to stimuli. A stimulus is a change in the environment that causes a response. Example: light, heat, sound, touch, smell.

Multicellular animals need a system to receive information, process it and give a response. Two systems work together:

- **Nervous system:** Fast, electrical and chemical signals through nerves. Short lived response.
- **Endocrine system:** Slow, chemical signals through hormones in blood. Long lasting response.

Plants do not have a nervous system. They coordinate responses through chemical signals and changes in growth or cell water content.

Structure of Neuron

Neuron

The structural and functional unit of the nervous system. It carries messages in the form of electrical impulses.

A neuron has three main parts:

- **1. Dendrites:** Branch like structures that receive signals from other neurons or receptors. They are at the receiving end.
- **2. Cell body (Cyton):** Contains the nucleus and cytoplasm. It processes the signal.
- **3. Axon:** Long fibre that carries the impulse away from the cell body towards the next neuron or muscle. It is covered by a myelin sheath in many neurons. The myelin sheath insulates the axon and makes impulse travel faster.

At the end of the axon there are nerve endings. The gap between two neurons is called the synapse.

Synapse

The tiny gap between the nerve ending of one neuron and the dendrite of the next neuron, where the impulse is passed by chemicals.

There are three types of neurons: sensory neurons carry signals from sense organs to the brain and spinal cord; motor neurons carry signals from the brain and spinal cord to muscles and glands; relay or association neurons connect sensory and motor neurons inside the central nervous system.

How Nerve Impulse Travels

Receptors in sense organs detect a stimulus. Example: photoreceptors in the eye detect light, gustatory receptors in the tongue detect taste, olfactory receptors in the nose detect smell, phonoreceptors in the ear detect sound, and thermoreceptors in the skin detect heat.

When a receptor is stimulated, it creates an electrical impulse. This impulse travels along the dendrite to the cell body and then along the axon. At the nerve ending,

chemicals called neurotransmitters are released into the synapse. These chemicals cross the synapse and start a new impulse in the next neuron. In this way the impulse moves from one neuron to another until it reaches the brain or spinal cord.

Reflex Action and Reflex Arc

Reflex Action

A sudden, automatic and involuntary response to a stimulus, without conscious thought. It protects the body from harm.

Example: pulling your hand away when you touch a hot pan, blinking when something comes near your eye, sneezing, coughing, knee jerk.

Reflex Arc

The pathway taken by a nerve impulse during a reflex action, from receptor to effector.

Steps of a reflex arc:

- Receptor detects the stimulus. For example, heat receptors in the skin.
- Sensory neuron carries the impulse to the spinal cord.
- Relay neuron inside the spinal cord processes the signal.
- Motor neuron carries the response signal from the spinal cord.
- Effector (muscle or gland) performs the response. The hand muscle contracts and pulls the hand back.

In a withdrawal reflex, the spinal cord coordinates the immediate response, making it very fast; the brain is also informed, so we become aware of it. Some other reflexes are coordinated by centres in the brain.

Worked Example

What is the difference between reflex action and voluntary action?

Reflex action is involuntary, fast and controlled by the spinal cord. Example: withdrawing hand from a hot object. Voluntary action is done with conscious thought and is controlled by the brain. Example: picking up a book. Reflex action is automatic, voluntary action is by choice.

Human Brain

The brain is the main coordinating centre of the body. It receives information from all sense organs, processes it and sends instructions. It is the most important part of the central nervous system.

The brain has three major parts:

1. Forebrain. It is the largest part. It has the cerebrum, thalamus and hypothalamus.

- **Cerebrum:** It is the thinking part. It controls voluntary actions like walking, speaking and writing. It is the site of intelligence, memory and reasoning. Different areas of cerebrum control different functions like sight, hearing and taste.
- **Thalamus:** Relays sensory information to the cerebrum.
- **Hypothalamus:** Controls hunger, thirst, body temperature and hormonal secretion from the pituitary gland.

2. Midbrain. It connects the forebrain to the hindbrain. It controls reflexes of the eyes and ears. It also controls involuntary movements.

3. Hindbrain. It has three parts:

- **Cerebellum:** Controls balance, posture and coordination of voluntary movements. It helps us to walk in a straight line and to ride a cycle.
- **Pons:** Controls breathing and connects different parts of the brain.
- **Medulla oblongata:** Controls involuntary actions like heartbeat, breathing, blood pressure, swallowing and vomiting. It connects the brain to the spinal cord.

The part of the brain that controls voluntary actions is the cerebrum. The part that controls involuntary actions is the medulla and the midbrain.

Protection of the Brain and Spinal Cord

- The brain sits inside the skull. The skull is a hard bony box that protects it.
- Inside the skull, the brain is covered by three membranes called meninges.
- Between the membranes there is cerebrospinal fluid. This fluid cushions the brain from shocks and provides nutrients.
- The spinal cord runs through the vertebral column. It also has meninges and cerebrospinal fluid for protection.

Coordination in Plants

Plants have no nervous system and no muscles. But they show movements. Plants respond to stimuli by growth movements. These movements are controlled by hormones.

There are two main types of plant movements:

1. Tropism (tropic movement). It is a directional growth movement towards or away from a stimulus. Growth is slow.

- **Phototropism:** Movement towards or away from light. Shoots bend towards light (positive phototropism). Roots bend away from light (negative phototropism).
- **Geotropism:** Movement in response to gravity. Roots grow down towards gravity (positive geotropism). Shoots grow up away from gravity (negative geotropism).
- **Hydrotropism:** Movement towards water. Roots grow towards water.
- **Chemotropism:** Movement towards chemicals. Pollen tube grows towards chemicals released by the ovary.
- **Thigmotropism:** Movement in response to touch. Tendrils of pea coil around a support.

2. Nastism (nastic movement). It is a non-directional movement that does not depend on the direction of stimulus. It is fast. Example: folding of leaves of Mimosa (touch me not) when touched. This is thigmonasty. Opening and closing of flowers in response to light is photonasty.

Worked Example

What is the difference between tropic and nastic movement?

Tropic movement is directional and depends on the direction of the stimulus. It is slow growth movement. Example: shoot bending towards light. Nastic movement is non-directional and does not depend on direction of stimulus. It is fast and often due to change in water content. Example: Mimosa leaves folding on touch.

Plant Hormones (Phytohormones)

Plant Hormone

A chemical substance produced in small amounts in one part of the plant and transported to another part where it controls growth and responses.

There are five major plant hormones:

- **1. Auxin:** Made at the tip of shoot and root. It helps in cell elongation. When light falls on one side of the shoot, auxin moves to the shaded side. The shaded side grows more and the shoot bends towards light. Auxin also helps in root growth and fruit formation.
- **2. Gibberellin:** Helps in stem elongation, seed germination and flowering. Farmers use it to increase stem length.
- **3. Cytokinin:** Helps in cell division. It is present in areas of rapid cell division like fruits and seeds. It delays ageing of leaves.
- **4. Abscissic Acid (ABA):** It is a growth inhibitor. It causes dormancy of seeds and wilting of leaves. It closes stomata during water stress.
- **5. Ethylene:** A gaseous hormone. It helps in ripening of fruits. It also causes falling of leaves and flowers.

Hormones in Animals

Hormone

A chemical messenger secreted by an endocrine gland in small amounts and carried by blood to a target organ where it controls a function.

Hormones are different from nerve impulses:

- Nerve impulses are fast and electrical, sent through neurons to a specific area. Hormones are slow and chemical, sent through blood to the whole body but act only on target organs.
- Nervous response is quick and short lived. Hormonal response is slow and long lasting.

Endocrine System in Humans

Humans have endocrine glands that release hormones directly into blood. They are ductless glands. Major glands are:

Gland	Hormone	Function
Hypothalamus	Releasing hormones	Controls pituitary gland
Pituitary (master gland)	Growth hormone	Controls growth. Less causes dwarfism, more causes gigantism
Pineal	Melatonin	Controls sleep cycle
Thyroid	Thyroxine (needs iodine)	Controls metabolism. Less causes goitre
Parathyroid	Parathormone	Controls calcium and phosphate balance
Thymus	Thymosin	Helps in immune system development
Adrenal	Adrenaline	Emergency hormone. Increases heartbeat and breathing in fear or anger

Gland	Hormone	Function
Pancreas	Insulin and Glucagon	Controls blood sugar. Insulin lowers blood sugar
Testes (in males)	Testosterone	Controls male characteristics and sperm formation
Ovaries (in females)	Oestrogen, Progesterone	Controls female characteristics, egg formation and pregnancy

Important points:

- Iodine is needed to make thyroxine. Lack of iodine causes goitre (swollen neck).
- Insufficient insulin action can cause diabetes mellitus, in which blood glucose remains abnormally high.
- Adrenal gland releases adrenaline in emergency. It prepares the body to fight or run.
- Pituitary is called the master gland because it controls other glands, but it is itself controlled by the hypothalamus.

Worked Example

A person has high blood sugar and sugar in urine. Which hormone and gland are involved?

Insulin from the pancreas controls blood sugar. When insulin is not enough, blood sugar stays high. This condition is called diabetes mellitus.

Feedback Mechanism

Hormone secretion is controlled by feedback mechanism. This keeps hormones in the right amount.

Feedback Mechanism

A system where the level of a hormone controls its further release. High level stops release and low level starts release.

Example: When blood sugar rises after eating, the pancreas releases more insulin. Insulin lowers blood sugar by helping cells take in glucose and converting extra glucose to glycogen in the liver. When blood sugar falls to normal, insulin secretion stops. When blood sugar falls too low, the pancreas releases glucagon which raises blood sugar. This is negative feedback.

Similarly, thyroxine controls its own release through feedback on the hypothalamus and pituitary.

Summary

- Animals coordinate through the nervous and endocrine systems. Plants respond through chemical signals and growth- or turgor-based movements.
- Neuron is the unit of the nervous system. It has dendrites, cell body and axon. Synapse is the gap between two neurons.
- Receptors detect stimuli. Impulse travels through sensory neuron to spinal cord or brain and response goes through motor neuron to effector.
- Reflex action is fast, automatic and controlled by the spinal cord. Reflex arc is the pathway of reflex action.
- Human brain has three parts: forebrain (cerebrum for thinking), midbrain (reflexes of eye and ear) and hindbrain (cerebellum for balance, pons for breathing, medulla for heartbeat and breathing).
- Brain is protected by skull, meninges and cerebrospinal fluid.
- Plant movements are tropic (directional growth like phototropism, geotropism) and nastic (non-directional like Mimosa folding).
- Plant hormones are auxin, gibberellin, cytokinin, abscisic acid and ethylene.
- Human endocrine glands include hypothalamus, pituitary, thyroid, parathyroid, thymus, adrenal, pancreas, testes and ovaries. Each releases a specific hormone.
- Hormone levels are controlled by feedback mechanism. Example: insulin and blood sugar.

Practice Questions

1. Draw a labelled diagram of a neuron and explain its three main parts.

2. What is a reflex arc? Trace the path of a reflex action when you touch a hot object.

3. What is the difference between sensory neuron and motor neuron?

4. Name the three parts of the human brain and give one function of each.

5. What is the difference between tropic and nastic movements? Give one example of each.

6. Name the five plant hormones and state one function of each.

7. Why is pituitary called the master gland? Which gland controls it?

8. Explain the feedback mechanism with the example of blood sugar control.